

Meta RL Overview

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Imagine an omnipresent assistant that understands who you are, anticipates what you need, and acts on your behalf by maintaining a single, evolving model of your life and acts on it over years. Going from an AI that responds when prompted to what we envision ("[Jarvis](#)") requires going well beyond scaling up current methods.

Evolution produced a learning algorithm—the human brain—that adapts to radically different environments within a single lifetime. A person trained as an ML researcher could have been a civil engineer instead; the underlying cognitive machinery is the same. The brain achieves this through a handful of architectural innovations: a hippocampal memory system for rapid, one-shot learning and selective retrieval; maintaining and updating states continuously; and a remarkable capacity to learn from sparse, costly feedback. These are precisely the capabilities that current AI lacks, and their absence defines the bottlenecks we must overcome.

This essay argues that these problems won't be solved by simply scaling compute, and outlines the genuinely new ideas required to make progress.

Why Scaling Won't Get Us There

Consider a first-time parent. Events unfold through a branching decision tree that unfolds over years, with each node depending on the resolution of previous uncertainties. Which pediatrician takes our insurance and has availability this month? The daycare waitlist says six months; should I put our name down at three places simultaneously? If so, which ones are near the office versus near home? How does that interact with the possibility that I might move this spring? Mother-in-law offered to help for two weeks; should I schedule that for right after birth or save it for when my spouse goes back to work? Each of these decisions depends on the other and they are laden with private preferences (how the parent feels about screen time or accepting help from family) that no model can infer from a single prompt.

Faced with any one of these questions, ChatGPT will answer by producing a competent and generic answer. But the parent's actual problem is not any single question—it is the web of interdependencies among them. GPT can answer "what should I look for in a daycare?" with encyclopedic thoroughness, but it cannot reason that this particular family should prioritize proximity to the suburban house they are considering, that the waitlist at Center A closes next Tuesday, and that putting down the deposit makes sense only if the move timeline firms up.

A human assistant, on the other hand, will not contribute by answering any single question but by maintaining a mental model of the entire situation. The human assistant remembers that the family is considering moving, and tracks which uncertainties have resolved (the pediatrician accepted new patients) and which remain open (the daycare waitlists). They notice when time-sensitive actions approach their deadlines and proactively flags them. Most importantly, they serve as the cognitive integration layer that connects every thread in the parent's life into a

coherent picture and reasons about what to do next: they recognize the daycare decision depends on the moving decision, which depends on the lease renewal deadline, which interacts with the budget after parental leave.

One might hope scaling would eventually bridge this gap. The bottleneck, however, is not the breadth of the model's knowledge, but rather its inability to maintain and reason over a particular user's state across time. This is not to say we won't see advances within the scaling paradigm: we'll soon get competent tool use and longer reasoning chains that lead to a capable "Claude with MCP access to your calendar". But these will not produce agents that operate in vast action spaces that change daily, proactively identifying which dimensions matter to the user based on personal taste, financial situation, and family dynamics. No amount of pretraining data can teach a model the daycare waitlist status for a specific family in a specific city. These are private, dynamic, context-dependent facts that exist only in the user's life and can only be learned through *interaction*. Scaling produces ever more knowledgeable encyclopedias, but it won't produce a system that tracks the state of your world and reasons about what to do next.

Intellectual Bottlenecks

Today's training regime does not equip models to deal with ambiguous situations that are long-horizon, context-limited, and where interactions are genuinely costly. Personalization is the canonical instance of this problem. Each interaction with a user reveals only a sliver of their preferences, and asking the wrong question wastes a scarce resource—the user's attention and patience. The system must learn to actively explore and resolve uncertainty, not merely retrieve and generate.

We want to build an AI system that maintains a persistent, structured representation of the user's life—their goals, constraints, uncertainties, and the dependencies among them—and updates this representation continuously as new information arrives. Where the current paradigm forces the user to serve as the sole integrating intelligence across a hundred chat threads, a state-aware system takes over that burden. It knows that the daycare question is connected to the potential move, which is connected to the commute calculation. It tracks which uncertainties have been resolved and which remain open. It prioritizes actions that are time-sensitive or that gate downstream decisions.

We delineate three major intellectual bottlenecks towards our vision, each of which will be addressed by a corresponding pillar of our proposed research agenda.

Bottleneck 1: Memory

LLMs have no episodic memory. The context window functions like working memory but it is limited and transient. There is no mechanism to form, consolidate, and selectively retrieve new memories after training. Retrieval augmented generation (RAG) is an attempt to bolt on a separate database, but the architectural mismatch is deep. No gradient signal flows from task performance back through storage and retrieval decisions. The model cannot learn what to

store or when to recall it, because the retrieval system is trained independently of the reasoning system.

We need a memory architecture where storage, consolidation, and retrieval are all end-to-end differentiable—so the system can learn, from the downstream personalization objective, which memories to form, how to organize them, and when to surface them. Beyond retrieval, the system must also learn to consolidate and forget. Over repeated interactions, the raw episodic record becomes unwieldy. The system must learn to compress experiences into compact, reusable semantic representations—extracting that this family prioritizes outdoor time and short commutes rather than storing the verbatim transcript of the conversation where this was discussed. This transition from episodic to semantic memory must be learned end-to-end, so the system discovers what level of abstraction serves downstream decision-making best.

Bottleneck 2: Data Efficiency

We need reinforcement learning algorithms that are dramatically more sample-efficient than what we have today. The current RL methods that underpin post-training—REINFORCE-style policy gradients and their descendants like [DeepSeek's GRPO](#)—are fundamentally zeroth-order: they treat the agent's entire reasoning process as a black box, observe only a scalar reward at the end, and use it to nudge trillions of parameters. This is extraordinarily slow and wasteful.¹ The data efficiency of these estimators deteriorates exponentially with the planning horizon (the classical “curse of horizon” in RL), which is fatal for personalization problems with a horizon of months to years! We need learning algorithms that exploit the known structure of the agent's planning problem that don't require trillions of trajectories to learn.

Bottleneck 3: Ambiguity and Active Exploration²

A central challenge in personalization is its partially observable nature. The user's preferences, context, and goals constitute a latent state that can only be inferred through strategic questioning and observation. This necessitates a system that maintains a belief over the latent state, reasons about which information would most reduce uncertainty, and plans multi-step interactions accordingly. A new parent's “state” is high dimensional, with a few key relevant details: whether they will move this spring, the daycare waitlist outcomes at three different centers, the mother-in-law's availability, the family's evolving budget after parental leave. A competent assistant must i) recognize critical dimensions, ii) form beliefs, iii) update them as uncertainties resolve on specific timelines (the daycare acceptance letters, the lease renewal deadline), and iv) prioritize actions that are time-sensitive or that gate downstream decisions.

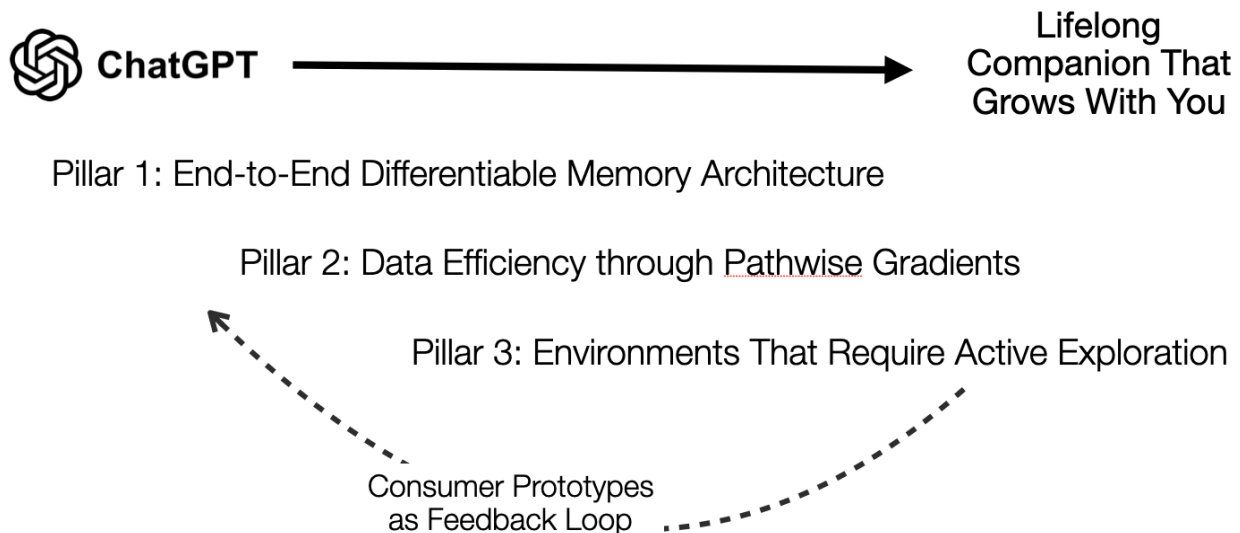
¹ Evolution is also a zeroth-order optimization method. It optimizes trillions of parameters (the genome) based on binary feedback (survival and reproduction). As a result, it is unbearably slow: it took 4 billion years to go from the first organic molecules to the current human brain.

² Tech researchers call this Adaptive Experimentation, RL researchers call this Active Exploration, Dave Blei calls this Bayesian Optimization. The most precise mathematical name for this is Bayesian Markov decision processes. I will use all of them interchangeably.

Note the stark contrast to current systems that treat each conversation turn as an independent prediction problem. This is because AI is trained on prompts that have a clear solution without ambiguity. The latest version of this is done on gourmet simulators of LinkedIn built by so-called “RL env” startups on which agents are trained. This is not only wasteful but non-sensical: humans do not learn to interact with the web by clicking on every button on LinkedIn! The unit economics of building such faithful virtual environments is prohibitive, and scaling is thus impossible due to large variable cost.

Proposed Approach: Learn Algorithms

Jarvis is best viewed as an RL engine acting in the world: a belief state over the user and environment, a memory system that updates online, and an adaptive experimentation engine that chooses what to learn next. Our agenda rests on a foundational thesis: meta-learn the RL algorithm, not the model. Rather than training ever-larger models on static datasets, we propose to train an *algorithm* so that it can rapidly adapt to new users, new contexts, and new tasks at deployment time. Just as evolution did not produce a brain hard-coded for any particular environment but rather a general-purpose learning machine, we aim to produce an agent equipped with the right inductive biases and learning mechanisms to personalize itself efficiently to each user. The research agenda has three interlocking pillars, each designed to directly resolve one of the bottlenecks identified above.



Pillar 1: End-to-End Differentiable Memory Architecture

Transformers serve as a capable prefrontal cortex, excelling at flexible reasoning over whatever is present in working memory. But it has no mechanism for hippocampal functions: encoding new experiences, consolidating them, and selectively retrieving them when contextually

relevant.³ RAG is the field's attempt to graft on this missing organ by retrieving based on surface similarity over frozen embeddings, with no gradient signal flowing from task performance back through storage and retrieval decisions. The two systems are trained separately, so the model cannot learn end-to-end what to store or when to recall it.

Our proposal is a fully differentiable memory module that interfaces directly with the transformer's attention layers, so that storage, consolidation, and retrieval all admit gradient flow. We take inspiration from the communication loop between cortex and hippocampus: representations (hidden states) are routed to hippocampal circuits for rapid binding, and retrieval reactivates stored content and projects it back into the cortex. That is, we wish to inject episodic memories directly into the same representational space used for ongoing reasoning. Our main insight is that there should be no format mismatch between memory and cognition; storage and retrieval operate in the same substrate.

One concrete way to replicate this integration is through cross-attention layers interleaved with the transformer's self-attention blocks. The memory module maintains a differentiable external memory of key-value pairs. Each memory layer performs three operations: a) read, where current hidden states query the memory via multi-head cross-attention; b) gate, where a learned parameters control how much retrieved content blends into the residual stream; and c) write, where output representations are committed to memory through attention-weighted updates, with a learned network determining what is worth storing. This is somewhat analogous to the hippocampal novelty signal that triggers long-term storage. Because every operation is differentiable, gradients from the task loss flow back through retrieval (the system learns what to recall) and through storage (the system learns what to remember).

Memory consolidation: episodic to semantic Since memory grows with every interaction, the system must periodically compress it to preserve high-value entries while merging low-value ones. (Just like a hippocampus during we sleep.) We *train* this consolidation process end-to-end on the personalization objective, so that a natural transition from episodic to semantic memory emerges: the verbatim record of the parent is eventually compressed into a compact preference representation, freeing memory capacity while preserving decision-relevant information. Importantly, retrieval operates not by keyword overlap but by learned relevance—recognizing, for instance, that the resolution of the moving uncertainty is the moment when previously stored contingency plans about daycare become actionable.

Pillar 2: Data Efficiency through Pathwise Gradients

Current post-training methods for language agents are zeroth-order. Like evolution, they use binary outcomes to figure out how to update the LLM's trillions of parameters. Our key insight is that the agent's planning problem has exploitable structure that makes first-order methods possible. Consider an agent who maintains implicit beliefs over a parent's latent state and selects actions to reduce uncertainty. The dynamics of this process are not arbitrary—they are

³ Whether biological motivations are useful is a hotly debated topic in AI. I personally take on an operational view that while it may be useful for inspiration, ultimately relevant solutions are borne out of mathematical algorithms and robust engineering, all validated through rigorous empirical tests.

governed by Bayes' rule, which updates the posterior from the prior belief and the new observation. Even the reward, measuring how much uncertainty was reduced, is a differentiable function of the belief state. Together, this means the entire inner loop of the agent's adaptive experimentation admits a computational graph through which gradients can flow.

We formalize this as a Markov decision process over belief states.⁴ At each step, the agent selects an action (a question to ask, an item to recommend, a piece of information to surface), observes the user's response, and updates its belief via Bayes' rule. The reward at each step measures the value of information gained—for instance, the reduction in entropy of the posterior over the user's latent preferences. Because both the transition function (posterior update) and the reward function (entropy reduction) are differentiable operations on continuous belief representations, we can backpropagate the cumulative reward signal directly through the agent's trajectory of decisions, rather than relying on Monte Carlo estimates of the return.

The key technical challenge here is that the agent's actions are generated in discrete token space, which breaks the gradient chain. We address this through careful smoothing approximations: replacing the hard argmax of token sampling with a temperature-controlled softmax, and using the [straight-through estimator](#) to maintain gradient flow through discrete choices. The resulting pathwise gradient estimator has dramatically lower variance than REINFORCE. In our prior work on smoothed autodifferentiation for long-horizon RL problems⁵, pathwise methods achieved 1000x improvements in sample efficiency over state-of-the-art RL methods in long-horizon settings. The variance reduction here is *transformative*, because the variance of Monte Carlo estimators grows exponentially with the planning horizon (the classical “curse of horizon” in RL), while pathwise estimators exploit the structure of the dynamics to maintain tractable variance regardless of horizon length.

Pillar 3: Environments That Require Active Exploration

We need training environments that are simple enough to build at scale but complex enough to require genuine exploration. Our capability target is identifying which dimensions of a problem are important and determining which information would resolve future uncertainty. We want to build a curriculum of environments that demand active exploration under ambiguity, ranging from simple browser-based tasks to rich simulations of everyday life for diverse users. These environments must contain hidden user preferences and changing external conditions, so the agent must learn by interaction, not distillation. But these environments do not need to be a faithful replica of reality if the goal is to teach agents to be effective at long-term planning (a “good RL algorithm”). The process of sequentially gathering information from webpages and interacting with the user generalizes well across webpages—we don't need to learn this on the most realistic implementation of LinkedIn.

⁴ [Adaptive Experimentation at Scale: A Computational Framework for Flexible Batches](#)
[Optimization-Driven Adaptive Experimentation](#)

⁵ [Differentiable Discrete Event Simulation for Queuing Network Control](#)
[A Planning Framework for Adaptive Labeling](#)

Going back to the new parent example, we will need to simulate users with different family structures, different geographies, different financial situations, and different (latent) preferences about parenting philosophy, work-life balance, and risk tolerance. The agent trained in this environment should learn not the answers to any specific parenting question, but the algorithm for figuring out what matters to a specific parent and acquiring that information efficiently. Our prior work on personalization benchmarks and adaptive elicitation environments⁶ equip us with hard-earned lessons toward scaling to the complexity and open-endedness required for general personalization.

Feedback loop: Instrumentation via Real-World Interactions

Finally, we need to run an “integration test” of how these components fit together: context-driven recall of goal-relevant memories, integrated seamlessly into ongoing reasoning, driving long-horizon personalization in uncertain environments. In our recent work on adaptive elicitation⁷, we introduced a framework that meta-learns a language model to simulate future observations and select questions that maximally reduce epistemic uncertainty about a latent entity. This work lays the foundations, but scaling it to the full complexity of real life requires significant work.

The final component is thus deliberately unconventional and remains to be developed further. We propose to build and iterate on prototypes, not with the goal of commercial success, but as a scientific instrument for measuring the generalization gap. Laboratory environments, no matter how carefully designed, cannot fully replicate the distribution shift between training and deployment. A real system with real users provides the ground truth needed to validate or falsify whether our algorithms generalize. This mirrors best practices in the applied sciences: you build instruments to see what you otherwise cannot. Our prototype is meant to expose the failure modes that no simulator can predict, creating a feedback loop between fundamental research and reality.

Closing Thoughts

Our agenda addresses the three intellectual bottlenecks we identified as an integrated system. The differentiable memory architecture (Pillar 1) gives the agent the capacity to maintain persistent, structured representations of a user’s life, directly resolving the memory bottleneck that makes current systems amnesic. The pathwise gradient methods (Pillar 2) provide the sample efficiency needed to learn from the sparse, costly feedback that real human interactions produce, overcoming the data efficiency bottleneck that makes zeroth-order methods hopeless at personalization timescales. And the ambiguity-rich training environments (Pillar 3) provide the

⁶ [Benchmarking In-context Experiential Learning Through Repeated Product Recommendations](#), [PersonalLLM: Tailoring LLMs to Individual Preferences \(HF dataset\)](#), [LLM Generated Persona is a Promise with a Catch \(HF dataset\)](#)

⁷ [Adaptive Elicitation of Latent Information Using Natural Language Active Exploration via Autoregressive Generation of Missing Data](#)

curriculum that teaches agents to explore strategically—to ask the right questions, to resolve the right uncertainties, and to time their actions to the user’s real-world deadlines—addressing the exploration bottleneck at the heart of personalization. Together, these pillars constitute the infrastructure needed to shift AI from the thread paradigm to the trajectory paradigm: from stateless, disposable conversations to a single, evolving model of the user’s life.

Foundation models have given us the raw material: broad world knowledge and linguistic competence. What remains are scientific breakthroughs that demand a principled approach combined with the engineering ambition to build real systems and the intellectual honesty to let empirical reality guide the theory. If we get this right, the result will be a fundamental change in how people navigate the complexity of modern life.